MASTER IN ASTROPHYSICS AND SPACE SCIENCE

TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

Introduction to Numerical Methods for Astrophysics

Fritz Ali Agildere

January 23, 2026

Overview

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$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \boldsymbol{u} \cdot \nabla Q$$

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  - $M > 3$**:** Hypersonic (ram pressure stripping, dissociation and ionization)

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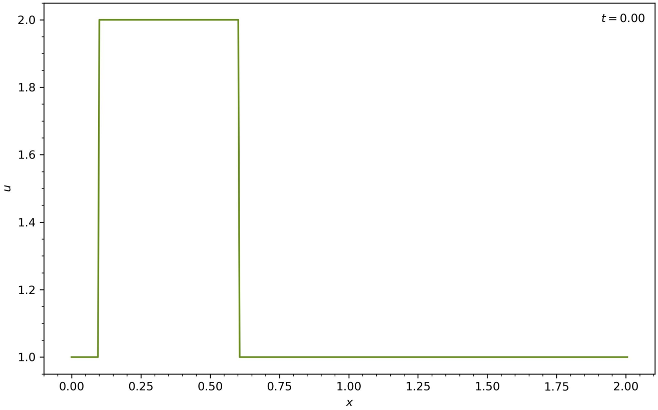

Linear Advection

- *Initialization:*
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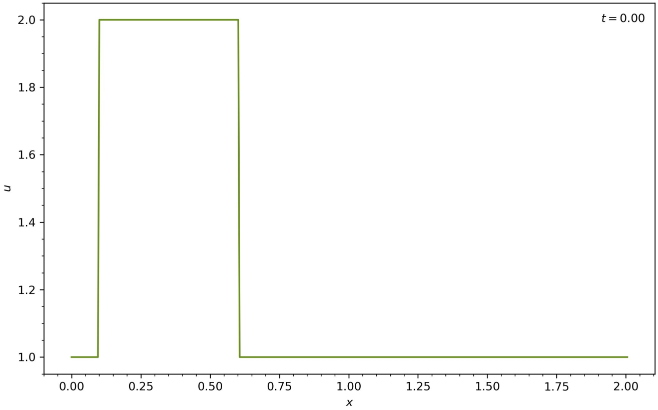

Linear Advection

- *Initialization:*
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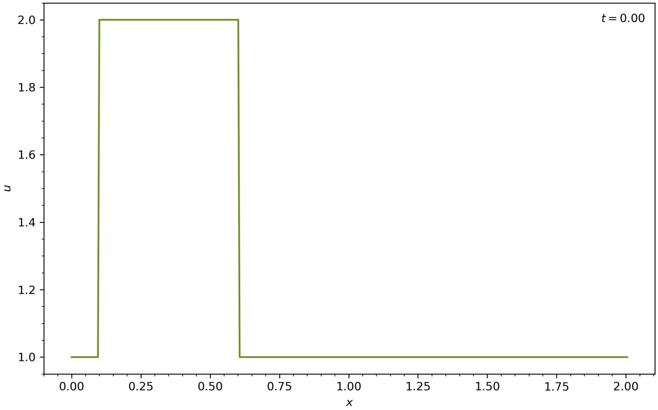
- *Boundaries:*



Linear Advection

- *Initialization:*
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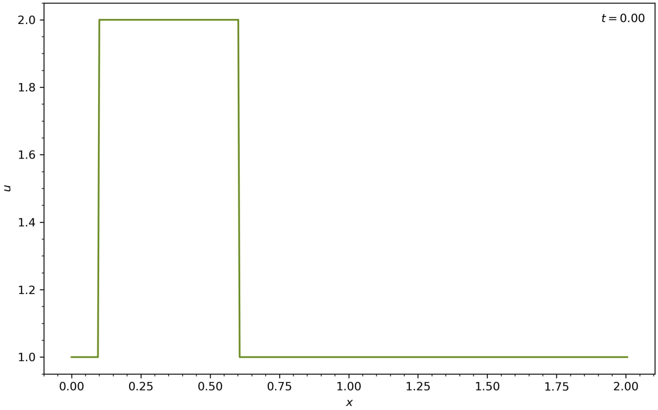
- *Boundaries:*
  - *Lower:* constant value



Linear Advection

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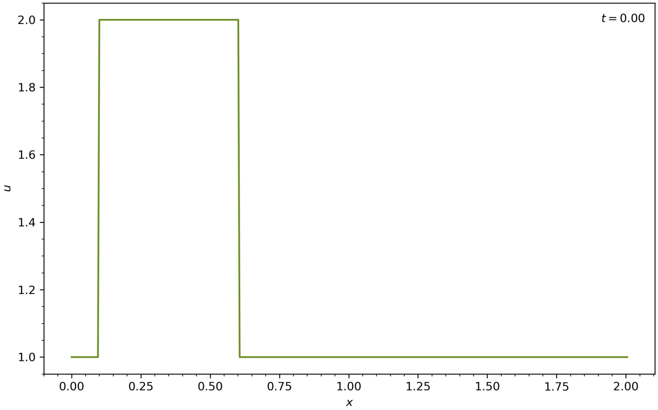

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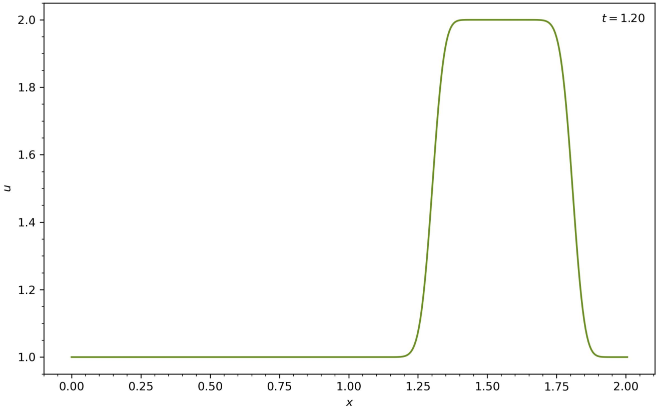

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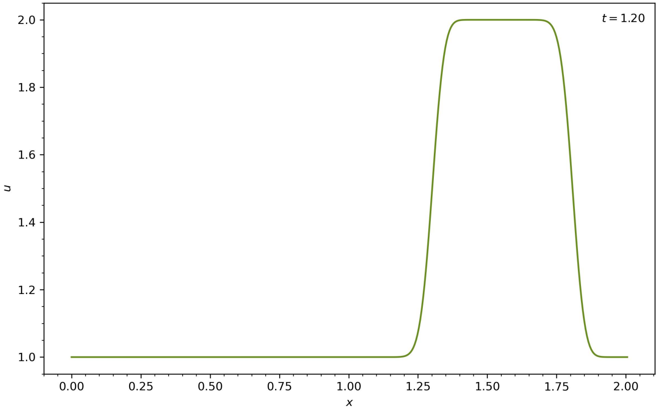

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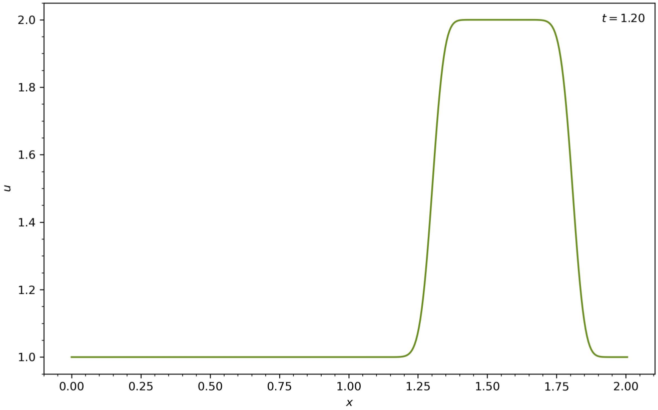
  - fixed profile
  - linear translation



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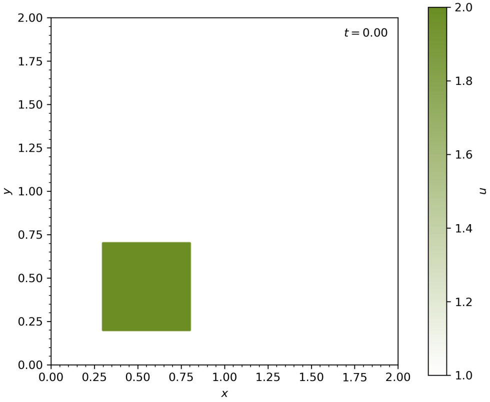

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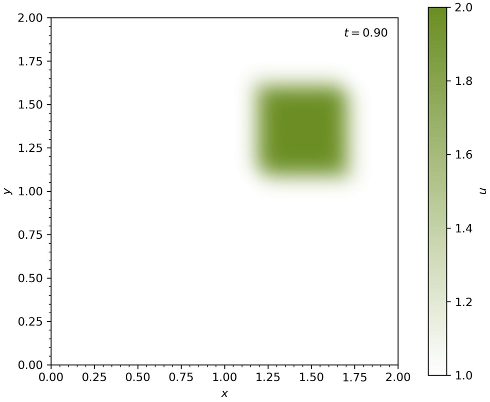

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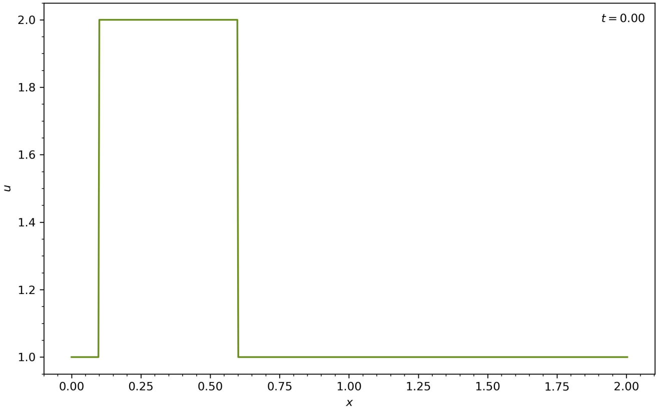

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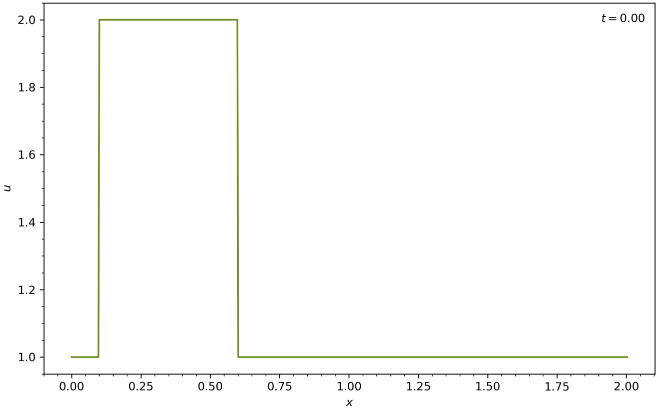

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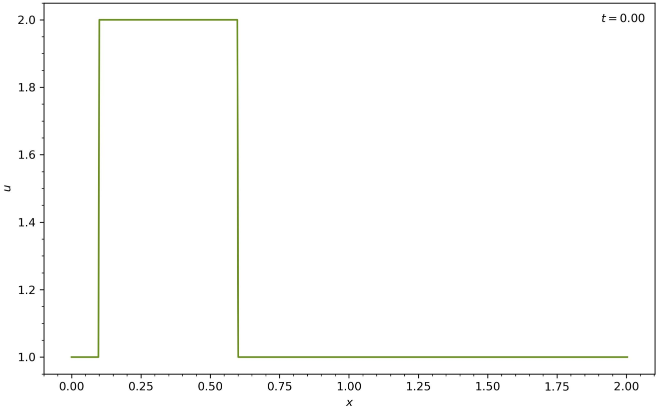

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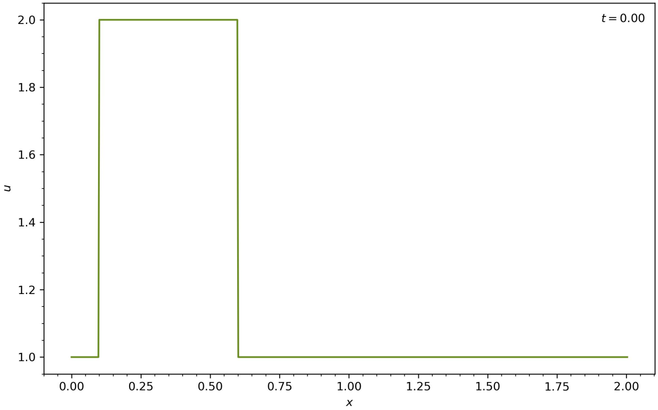

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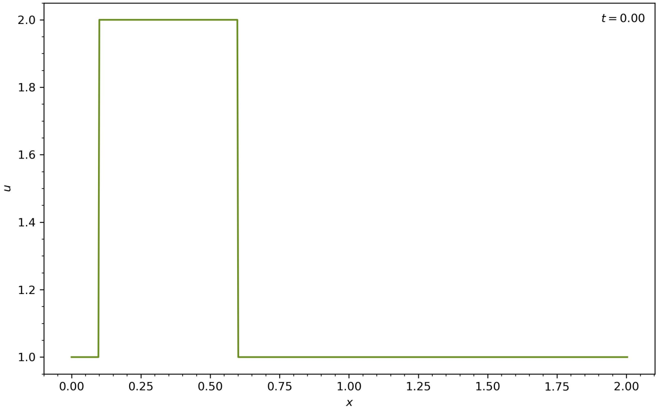

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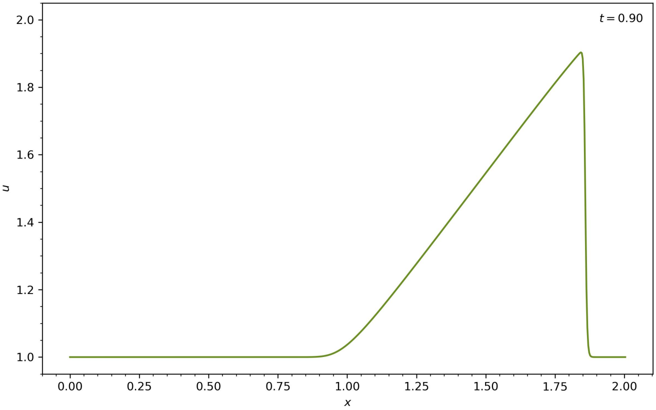

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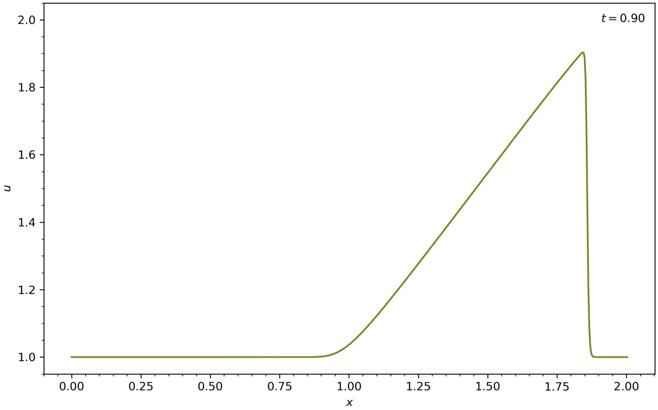

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- *Observation:*
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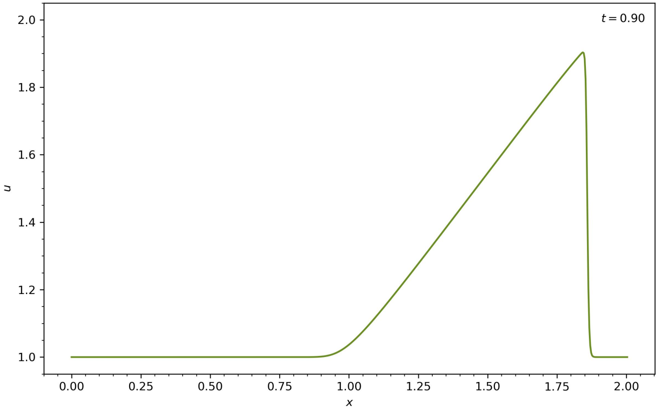
  - rarefaction tail



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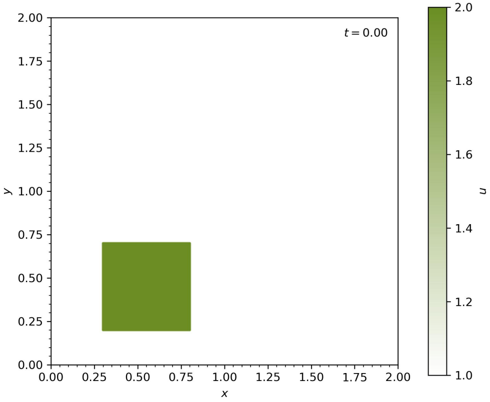

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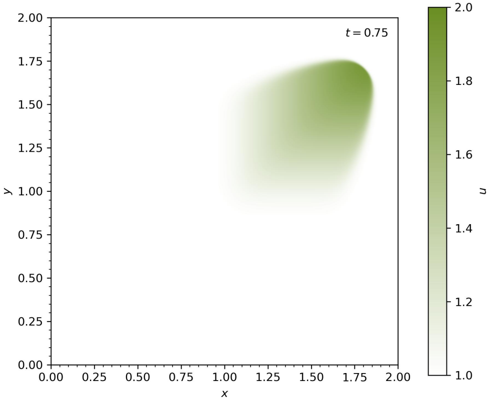

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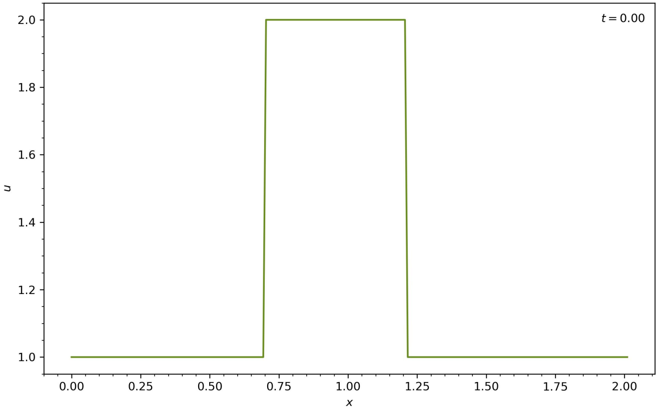

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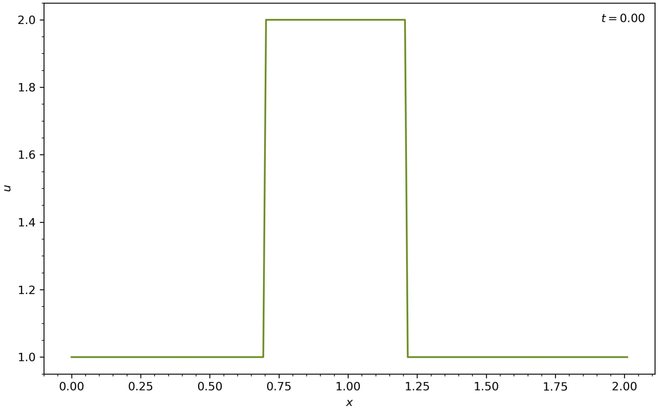

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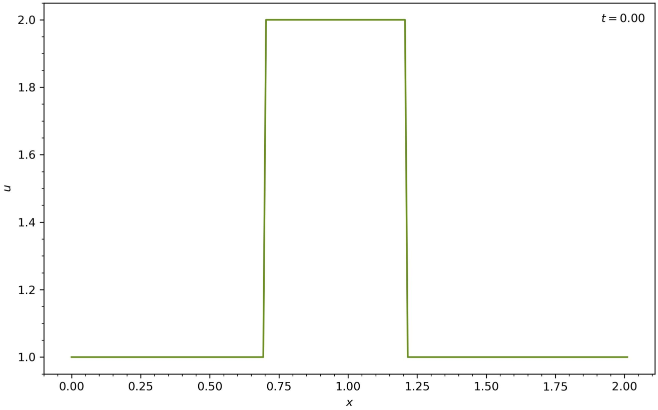

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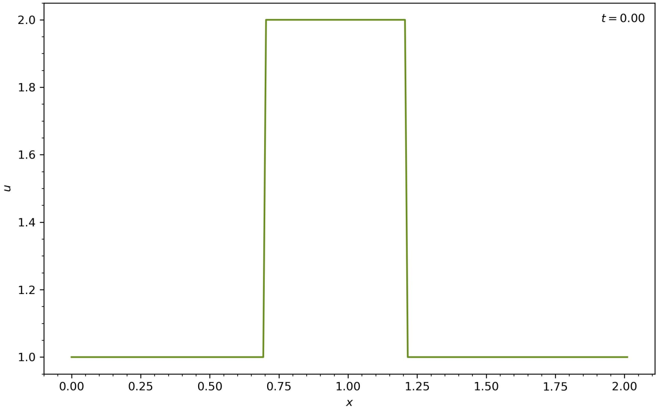

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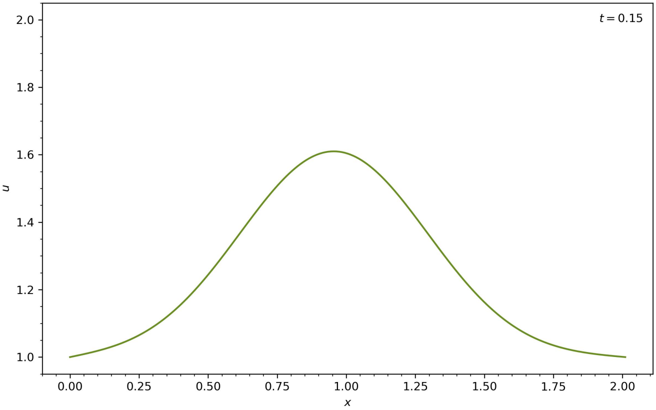

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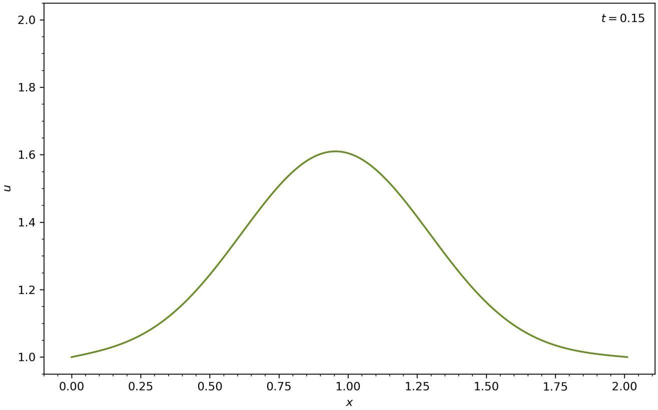

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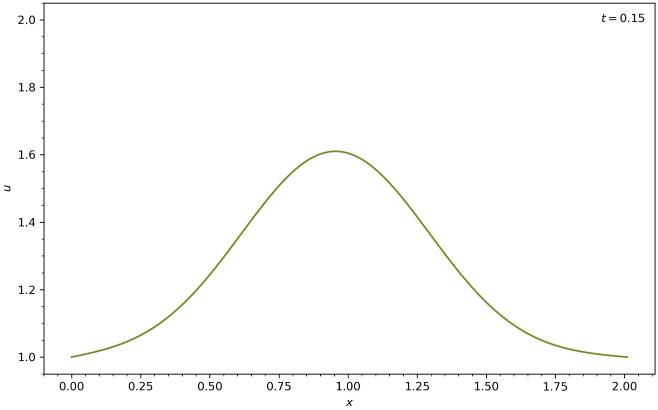

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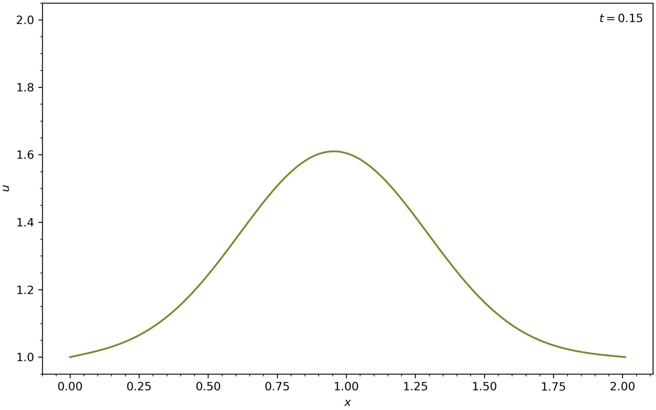
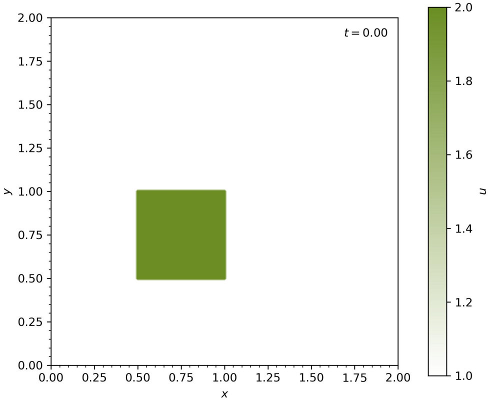
  - edge smoothing
  - spreading out



Diffusion

- *Initialization:*
  - square wave
- *Boundaries:*
  - *Both:* constant value
- *Observation:*
  - stationary decay
  - edge smoothing
  - spreading out



Diffusion

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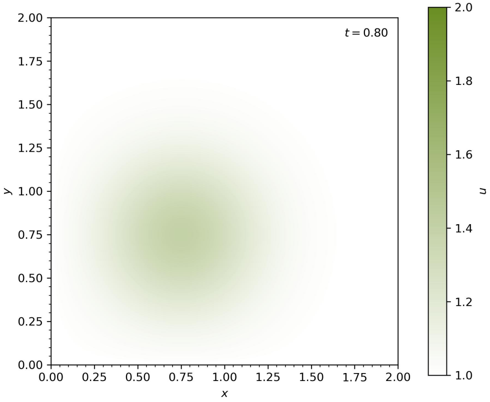
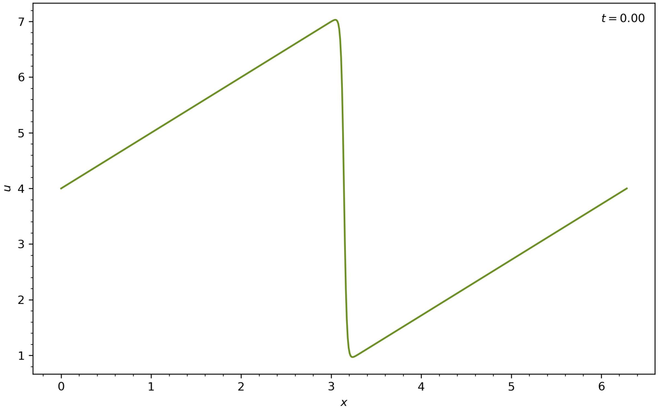
  - square wave
- *Boundaries:*
  - *Both:* constant value
- *Observation:*
  - stationary decay
  - edge smoothing
  - spreading out



Bateman-Burgers Equation

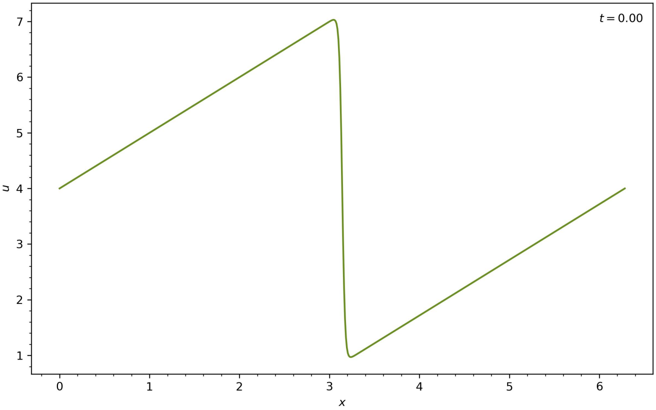
Bateman-Burgers Equation

- *Initialization:*



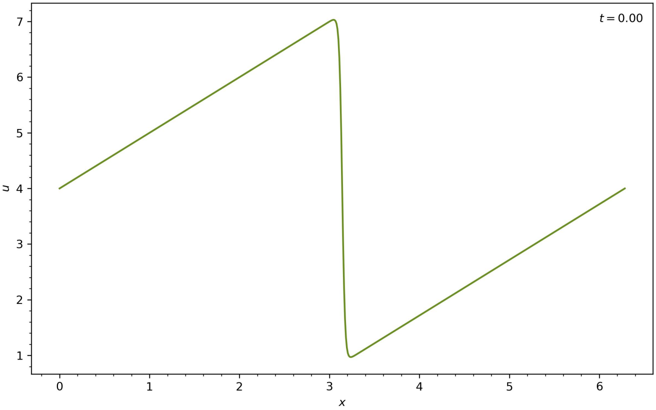
Bateman-Burgers Equation

- *Initialization:*
  - sawtooth wave



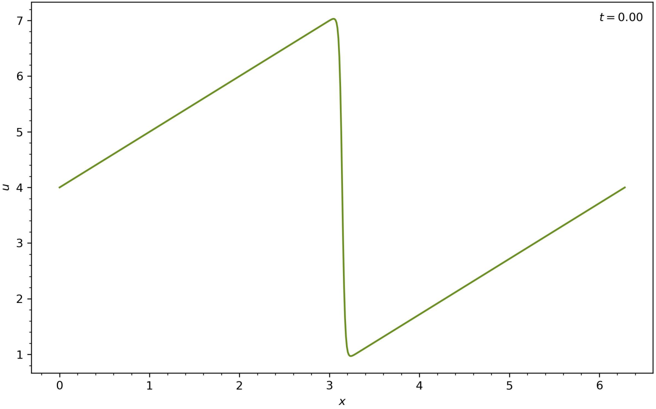
Bateman-Burgers Equation

- *Initialization:*
  - sawtooth wave
- *Boundaries:*



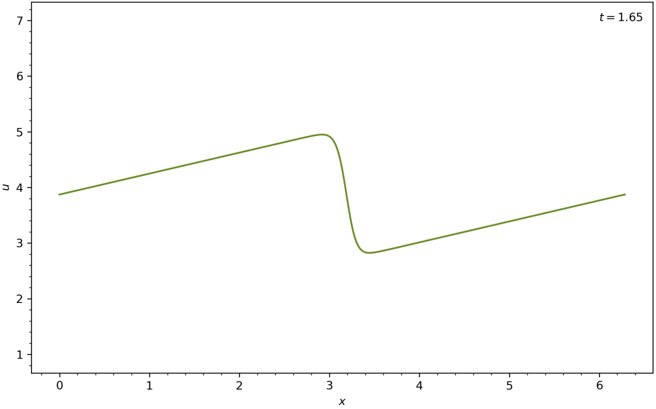
Bateman-Burgers Equation

- *Initialization:*
    - sawtooth wave
- *Boundaries:*
    - *Both:* periodic



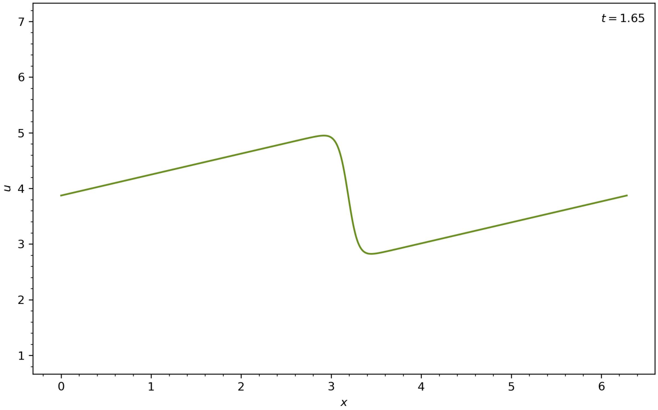
Bateman-Burgers Equation

- *Initialization:*
  - sawtooth wave
- *Boundaries:*
  - *Both:* periodic
- *Observation:*



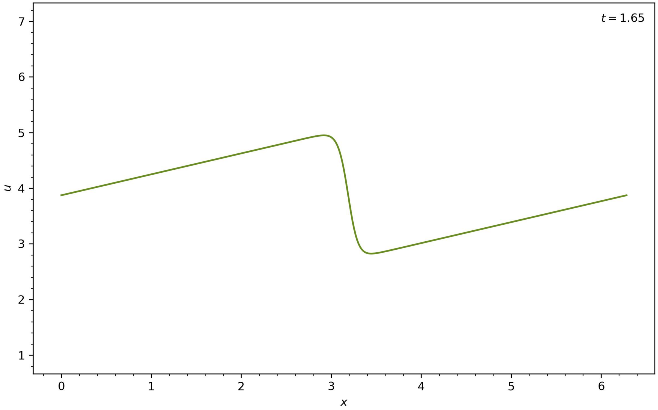
Bateman-Burgers Equation

- *Initialization:*
  - sawtooth wave
- *Boundaries:*
  - *Both:* periodic
- *Observation:*
  - nonlinear steepening



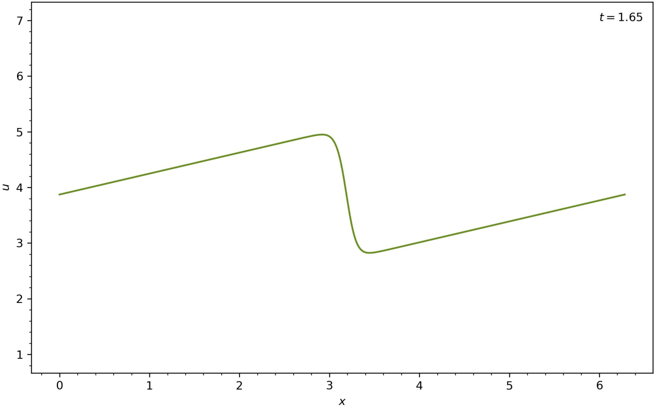
Bateman-Burgers Equation

- *Initialization:*
  - sawtooth wave
- *Boundaries:*
  - *Both:* periodic
- *Observation:*
  - nonlinear steepening
  - viscous smoothing



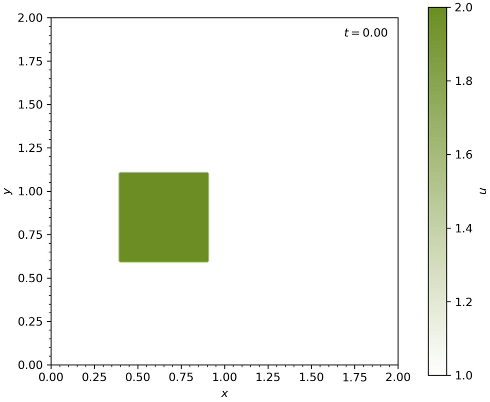
Bateman-Burgers Equation

- *Initialization:*
  - sawtooth wave
- *Boundaries:*
  - *Both:* periodic
- *Observation:*
  - nonlinear steepening
  - viscous smoothing
  - dissipating translation



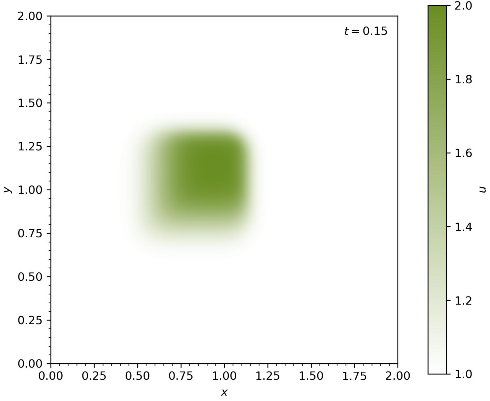
Bateman-Burgers Equation

- *Initialization:*
  - square wave
- *Boundaries:*
  - *Both:* constant value
- *Observation:*
  - nonlinear steepening
  - viscous smoothing
  - dissipating translation



Bateman-Burgers Equation

- *Initialization:*
  - square wave
- *Boundaries:*
  - *Both:* constant value
- *Observation:*
  - nonlinear steepening
  - viscous smoothing
  - dissipating translation



Equilibrium Search

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- *Laplace Equation:* maximally smooth solution dictated solely by boundary conditions

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$$\nabla^2 p = 0$$

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$$\nabla^2 p = f(r)$$

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- *Laplace Equation:* maximally smooth solution dictated solely by boundary conditions

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- *Poisson Equation:* extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

  - *Gravity:* $\nabla^2 \Phi = 4\pi G \rho$

Equilibrium Search

- *Laplace Equation:* maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation:* extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

  - *Gravity:* $\nabla^2 \Phi = 4\pi G \rho$
  - *Electrostatics:* $\nabla^2 V = -\rho / \varepsilon$

Equilibrium Search

- *Laplace Equation:* maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation:* extension for steady state to accomodate internal sources and sinks

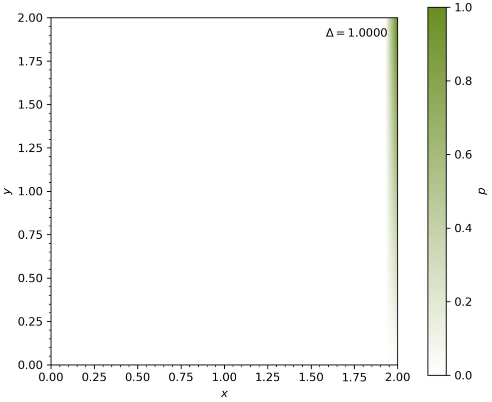
$$\nabla^2 p = f(r)$$

  - *Gravity:* $\nabla^2 \Phi = 4\pi G \rho$
  - *Electrostatics:* $\nabla^2 V = -\rho / \varepsilon$
  - *Pressure:* $\nabla^2 p = \rho \nabla \cdot \big((u \cdot \nabla) u\big)$

Laplace Equation

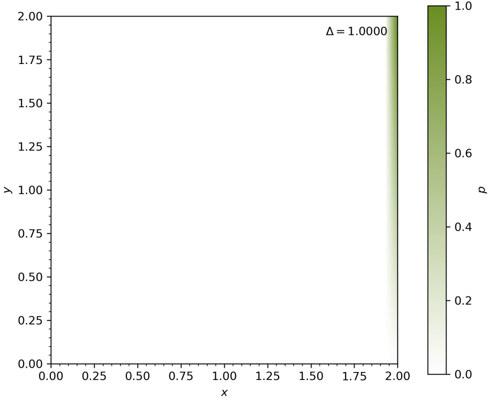
Laplace Equation

- *Initialization:*



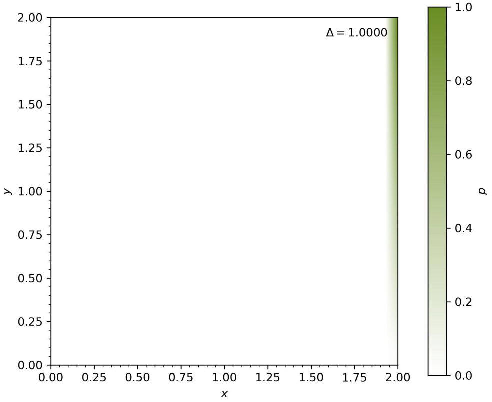
Laplace Equation

- *Initialization:*
  - empty field



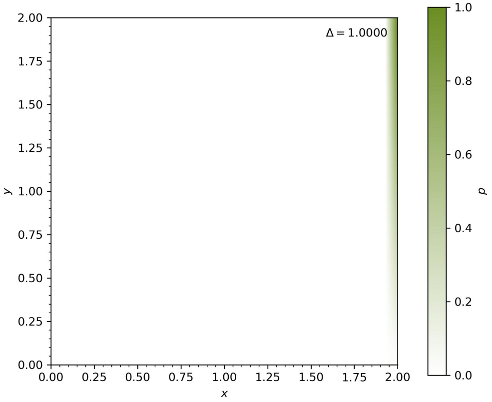
Laplace Equation

- *Initialization:*
  - empty field
- *Boundaries:*



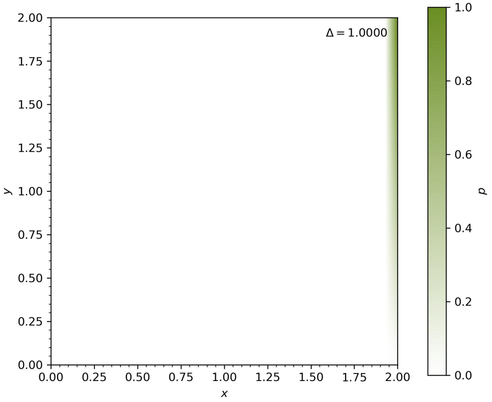
Laplace Equation

- *Initialization:*
  - empty field
- *Boundaries:*
  - *Left:* fixed zero



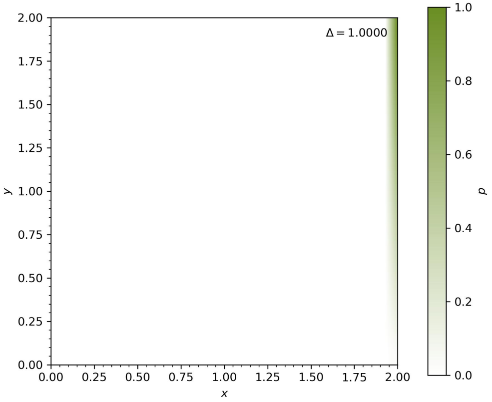
Laplace Equation

- *Initialization:*
  - empty field
- *Boundaries:*
  - *Left:* fixed zero
  - *Right:* linear slope



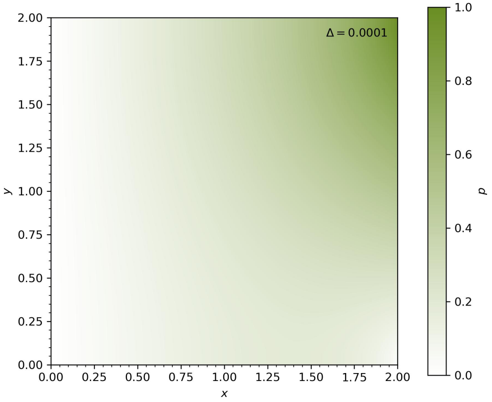
Laplace Equation

- *Initialization:*
  - empty field
- *Boundaries:*
  - *Left:* fixed zero
  - *Right:* linear slope
  - *Top & Bottom: no gradient*



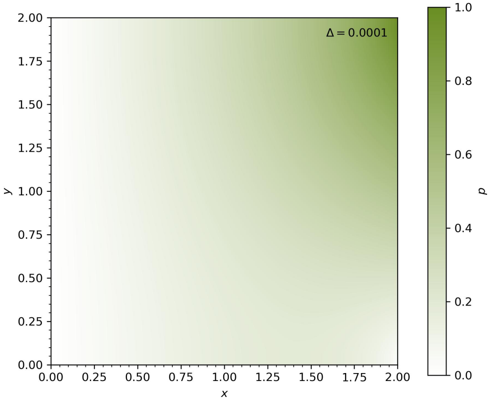
Laplace Equation

- *Initialization:*
  - empty field
- *Boundaries:*
  - *Left:* fixed zero
  - *Right:* linear slope
  - *Top & Bottom: no gradient*
- *Observation:*



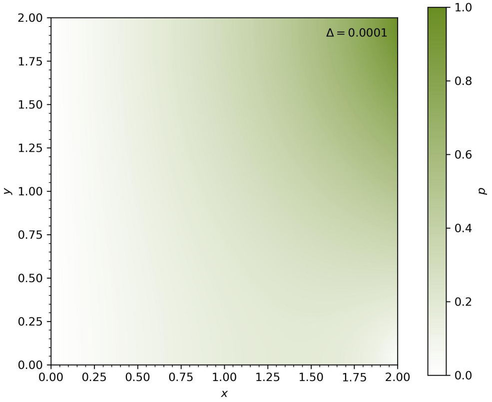
Laplace Equation

- *Initialization:*
    - empty field
- *Boundaries:*
    - *Left:* fixed zero
    - *Right:* linear slope
    - *Top & Bottom: no gradient*
- *Observation:*
    - iterative diffusion



Laplace Equation

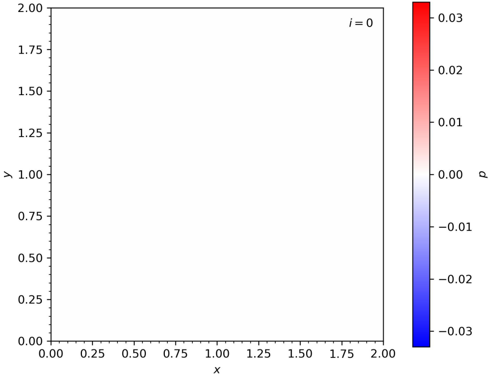
- *Initialization:*
  - empty field
- *Boundaries:*
  - *Left:* fixed zero
  - *Right:* linear slope
  - *Top & Bottom: no gradient*
- *Observation:*
  - iterative diffusion
  - smooth relaxation



Poisson Equation

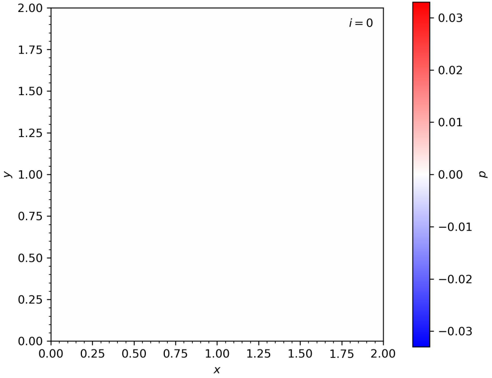
Poisson Equation

- *Initialization:*



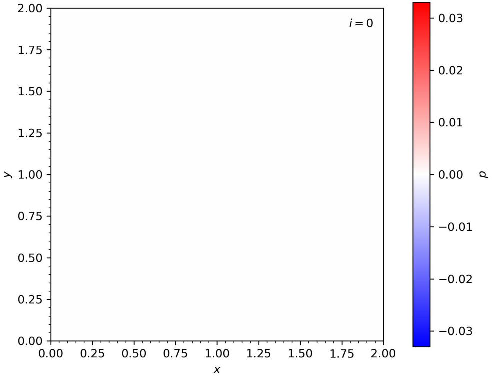
Poisson Equation

- *Initialization:*
    - empty field



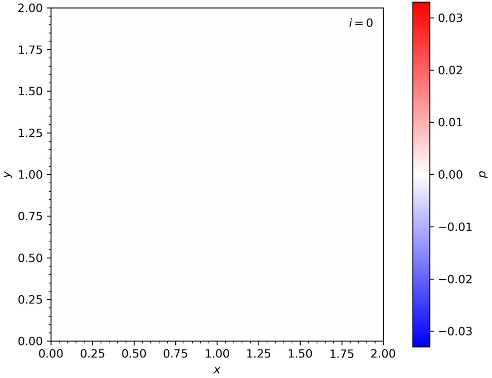
Poisson Equation

- *Initialization:*
  - empty field
  - positive source



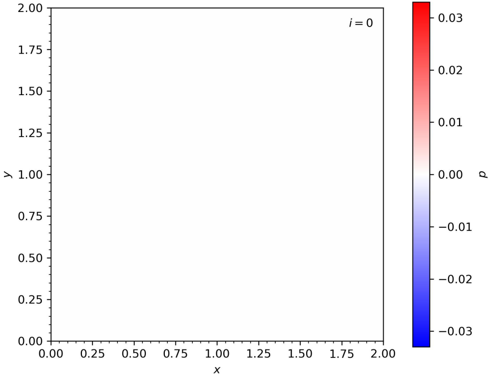
Poisson Equation

- *Initialization:*
  - empty field
  - positive source
  - negative sink



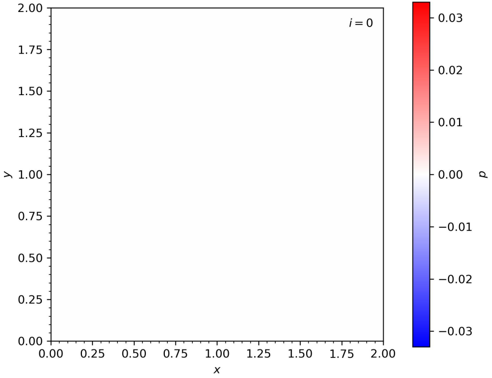
Poisson Equation

- *Initialization:*
  - empty field
  - positive source
  - negative sink
- *Boundaries:*



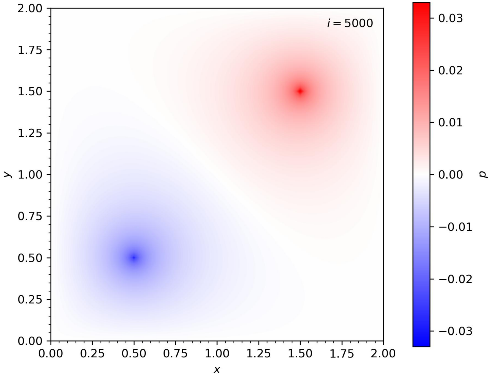
Poisson Equation

- *Initialization:*
  - empty field
  - positive source
  - negative sink
- *Boundaries:*
  - *All:* fixed zero



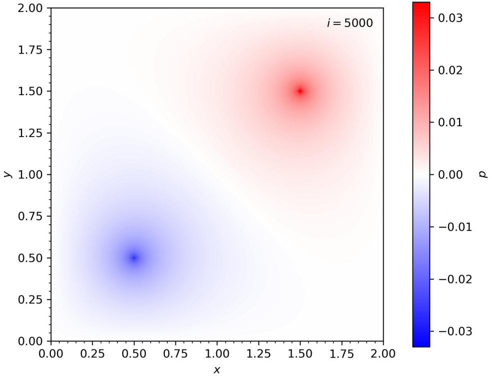
Poisson Equation

- *Initialization:*
  - empty field
  - positive source
  - negative sink
- *Boundaries:*
  - *All:* fixed zero
- *Observation:*



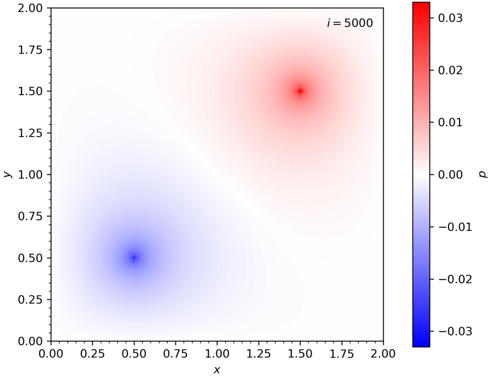
Poisson Equation

- *Initialization:*
  - empty field
  - positive source
  - negative sink
- *Boundaries:*
  - *All:* fixed zero
- *Observation:*
  - dipole field potential



Poisson Equation

- *Initialization:*
  - empty field
  - positive source
  - negative sink
- *Boundaries:*
  - *All:* fixed zero
- *Observation:*
  - dipole field potential
  - converging relaxation



Navier-Stokes Equations

Navier-Stokes Equations

- *Momentum Equation:* ensure $\nabla \cdot \boldsymbol{u} = 0$ by including ∇p to instantaneously cancel out any divergence

Navier-Stokes Equations

- *Momentum Equation:* ensure $\nabla \cdot \boldsymbol{u} = 0$ by including $\nabla \boldsymbol{p}$ to instantaneously cancel out any divergence

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\frac{1}{\rho}\nabla p + \nu \nabla^2 u$$

Navier-Stokes Equations

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- *Pressure Poisson:* require $\boldsymbol{p}$ to have relaxed to its equilibrium for all $\boldsymbol{t}$ before projecting

Navier-Stokes Equations

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$$\nabla^2 p = \rho\nabla \cdot \big((u \cdot \nabla)u\big)$$

Navier-Stokes Equations

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- *Pressure Poisson:* require $\boldsymbol{p}$ to have relaxed to its equilibrium for all $\boldsymbol{t}$ before projecting

$$\nabla^2 p = \rho \nabla \cdot \big((u \cdot \nabla)u\big)$$

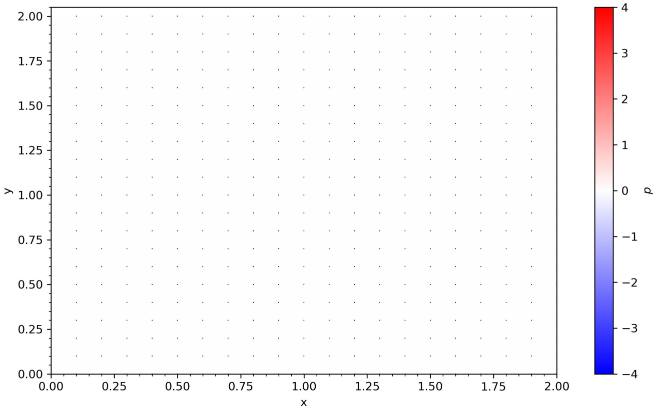
- fulfill at each step to accurately evolve initial configuration under given boundary conditions

Overview

Lid Driven Cavity Flow

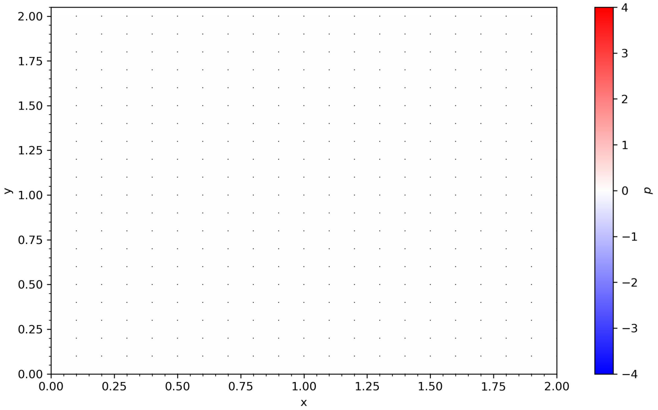
Lid Driven Cavity Flow

- *Initialization:*



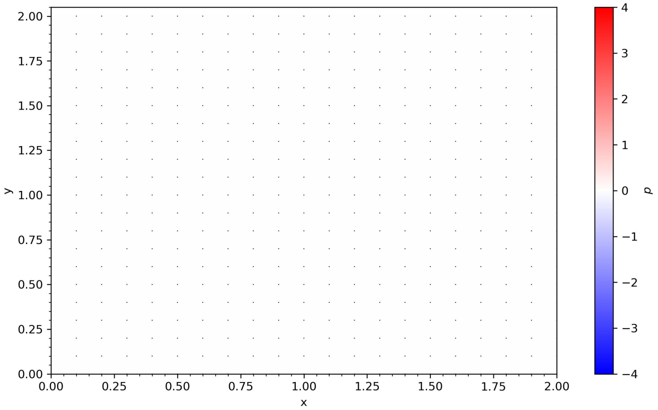
Lid Driven Cavity Flow

- *Initialization:*
    - vanishing velocity



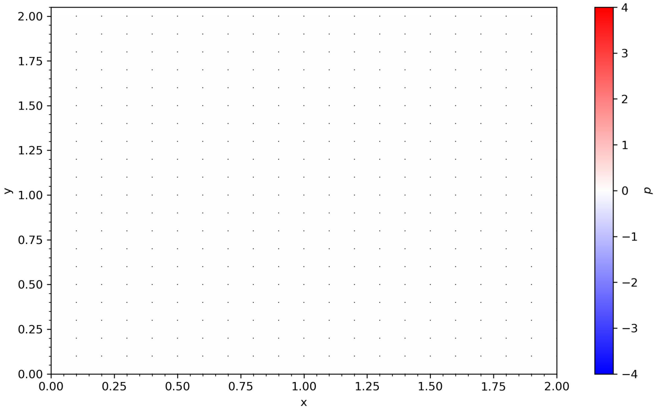
Lid Driven Cavity Flow

- *Initialization:*
  - vanishing velocity
  - zero pressure



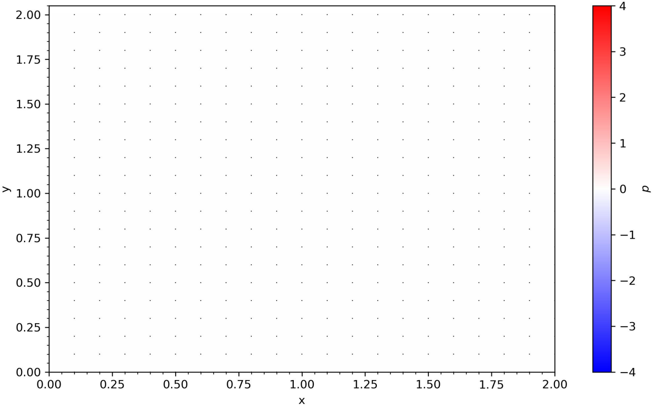
Lid Driven Cavity Flow

- *Initialization:*
    - vanishing velocity
    - zero pressure
- *Boundaries:*



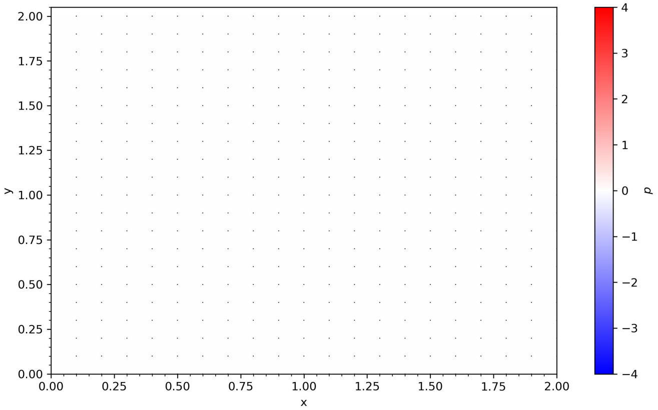
Lid Driven Cavity Flow

- *Initialization:*
  - vanishing velocity
  - zero pressure
- *Boundaries:*
  - *Top:* $u = 1$ & $\nabla p = 0$



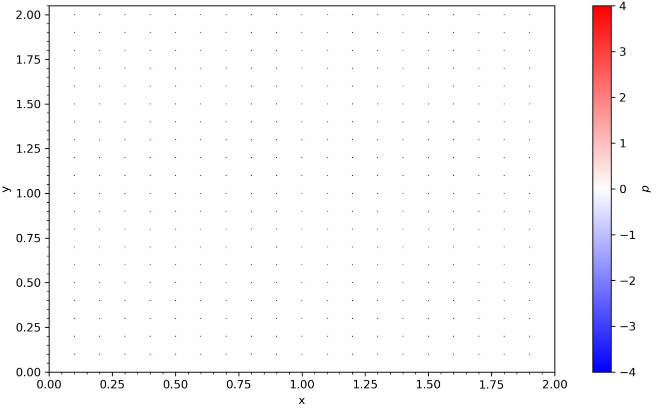
Lid Driven Cavity Flow

- *Initialization:*
  - vanishing velocity
  - zero pressure
- *Boundaries:*
  - *Top:* $\mathbf{u} = 1$ & $\nabla p = 0$
  - *Bottom:* $\mathbf{u} = 0$ & $p = 0$



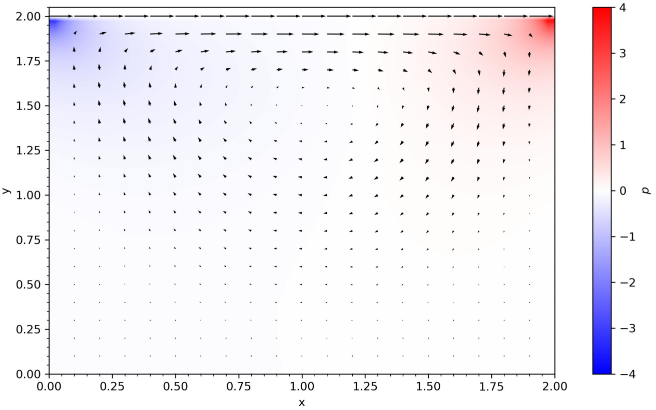
Lid Driven Cavity Flow

- *Initialization:*
  - vanishing velocity
  - zero pressure
- *Boundaries:*
  - *Top:* $\boldsymbol{u} = 1$ & $\nabla p = 0$
  - *Bottom:* $\boldsymbol{u} = 0$ & $p = 0$
  - *Left & Right:* $\boldsymbol{u} = 0$ & $\nabla p = 0$



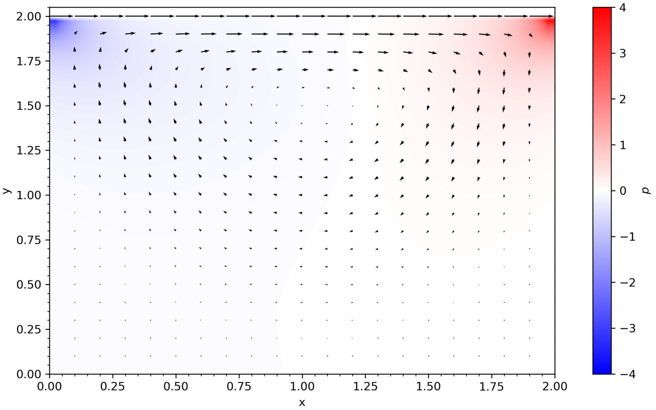
Lid Driven Cavity Flow

- *Initialization:*
  - vanishing velocity
  - zero pressure
- *Boundaries:*
  - *Top:* $\mathbf{u} = 1$ & $\nabla p = 0$
  - *Bottom:* $\mathbf{u} = 0$ & $p = 0$
  - *Left & Right:* $\mathbf{u} = 0$ & $\nabla p = 0$
- *Observation:*



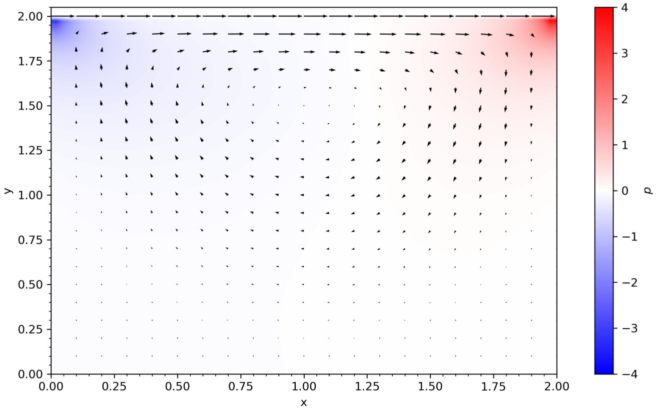
Lid Driven Cavity Flow

- *Initialization:*
  - vanishing velocity
  - zero pressure
- *Boundaries:*
  - *Top:* $\mathbf{u} = 1$ & $\nabla p = 0$
  - *Bottom:* $\mathbf{u} = 0$ & $p = 0$
  - *Left & Right:* $\mathbf{u} = 0$ & $\nabla p = 0$
- *Observation:*
  - vortex formation



Lid Driven Cavity Flow

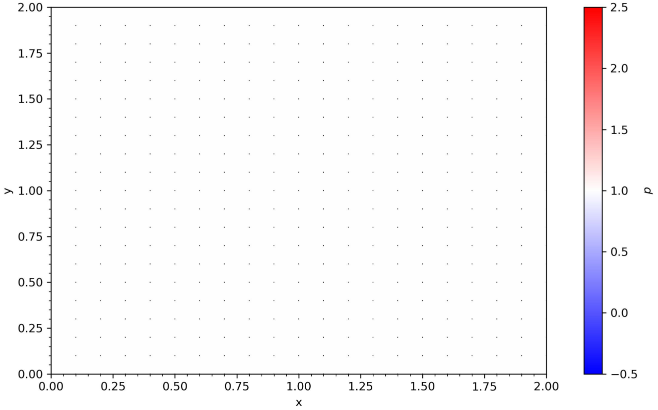
- *Initialization:*
  - vanishing velocity
  - zero pressure
- *Boundaries:*
  - *Top:* $\mathbf{u} = 1$ & $\nabla p = 0$
  - *Bottom:* $\mathbf{u} = 0$ & $p = 0$
  - *Left & Right:* $\mathbf{u} = 0$ & $\nabla p = 0$
- *Observation:*
  - vortex formation
  - pressure gradient



Pressure Driven Channel Flow

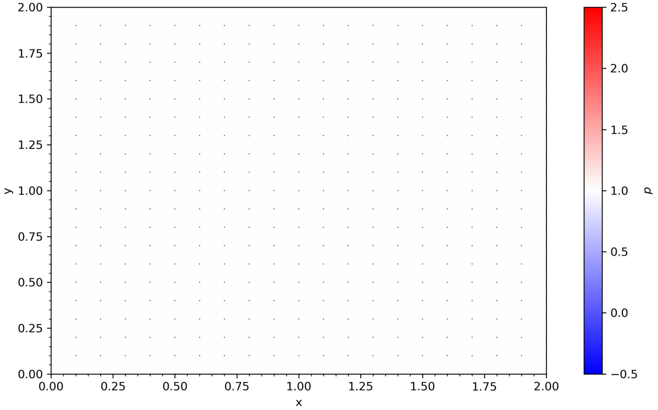
Pressure Driven Channel Flow

- *Initialization:*



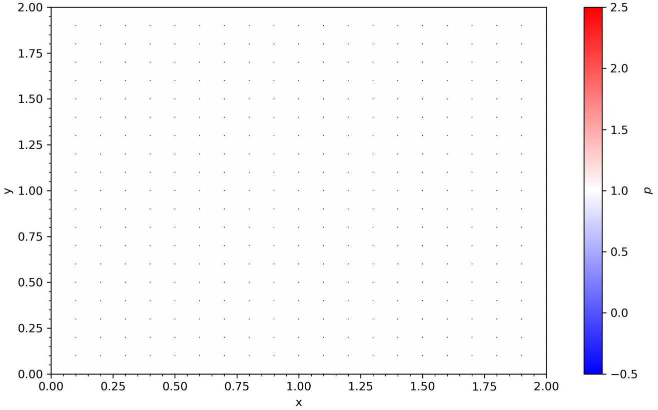
Pressure Driven Channel Flow

- *Initialization:*
  - vanishing velocity



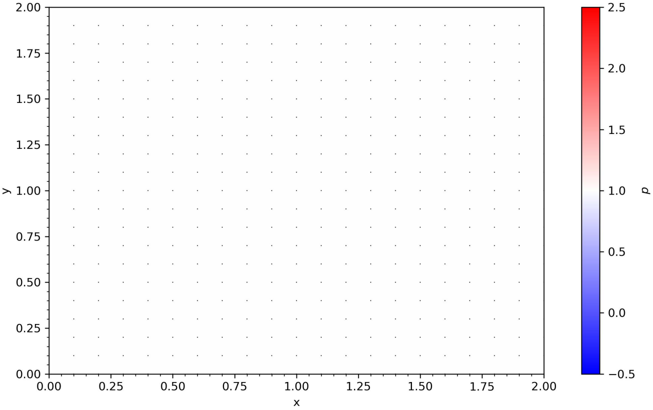
Pressure Driven Channel Flow

- *Initialization:*
  - vanishing velocity
  - unity pressure



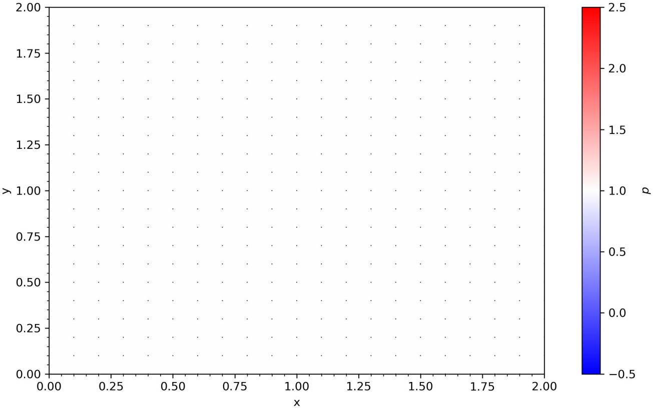
Pressure Driven Channel Flow

- *Initialization:*
  - vanishing velocity
  - unity pressure
  - *Poiseuille:* constant source



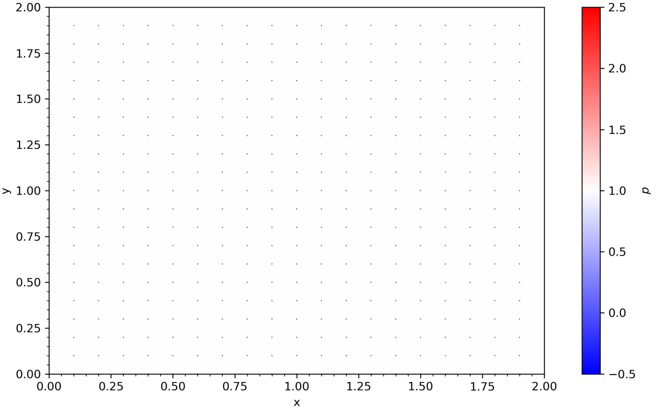
Pressure Driven Channel Flow

- *Initialization:*
  - vanishing velocity
  - unity pressure
  - *Poiseuille:* constant source
- *Boundaries:*



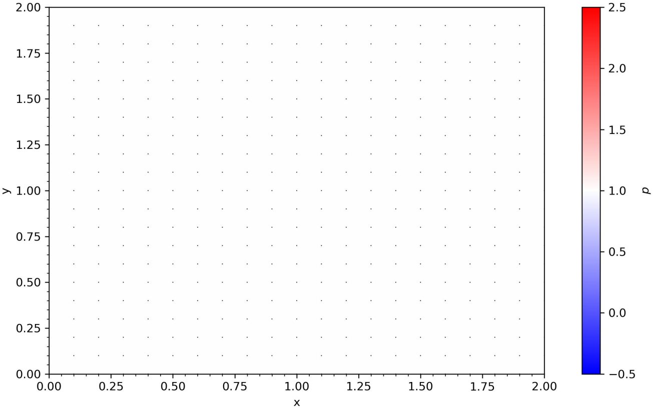
Pressure Driven Channel Flow

- *Initialization:*
    - vanishing velocity
    - unity pressure
    - *Poiseuille:* constant source
- *Boundaries:*
    - *Top & Bottom:* $\boldsymbol{u} = 0$ & $\nabla p = 0$



Pressure Driven Channel Flow

- *Initialization:*
  - vanishing velocity
  - unity pressure
  - *Poiseuille:* constant source
- *Boundaries:*
  - *Top & Bottom:* $\boldsymbol{u} = 0$ & $\nabla p = 0$
  - *Left & Right:* periodic



Pressure Driven Channel Flow

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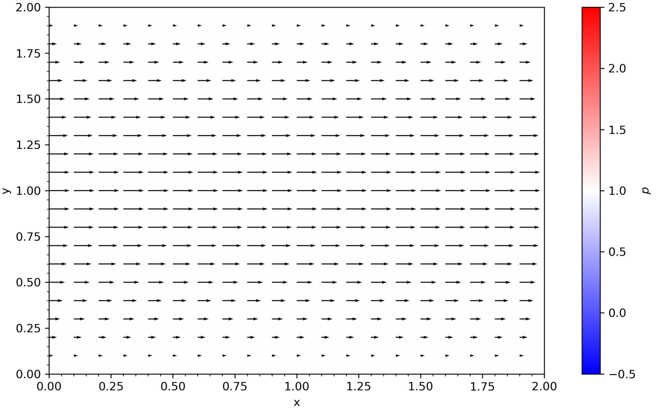

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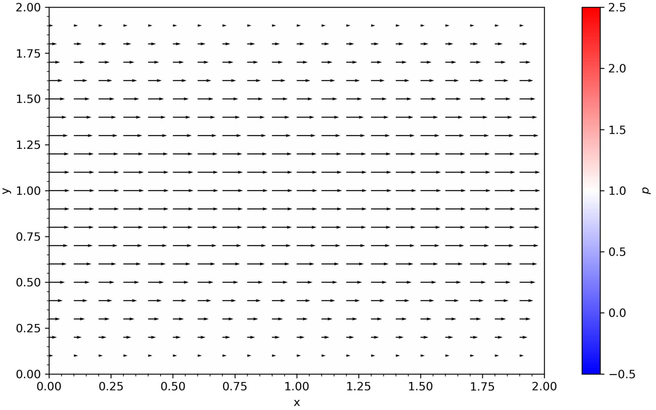

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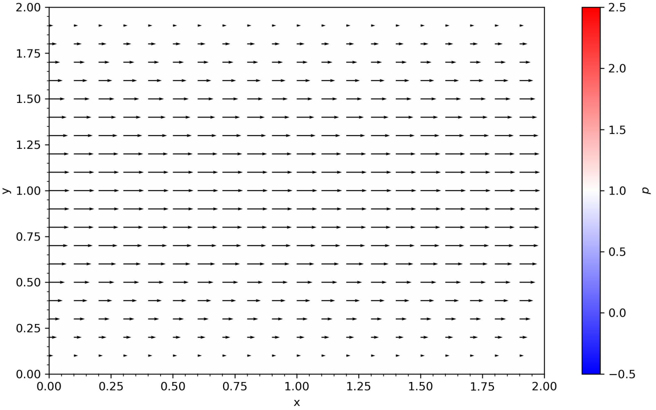

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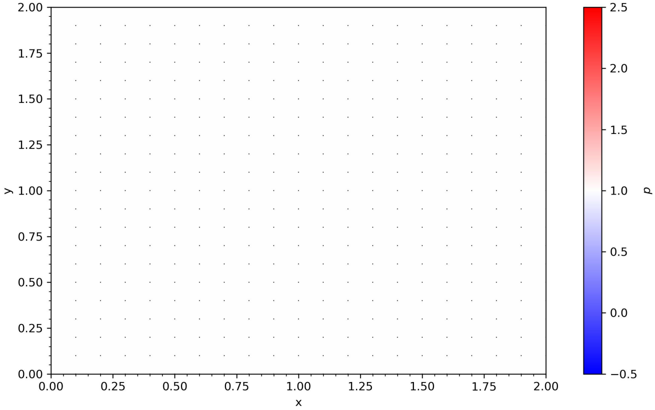

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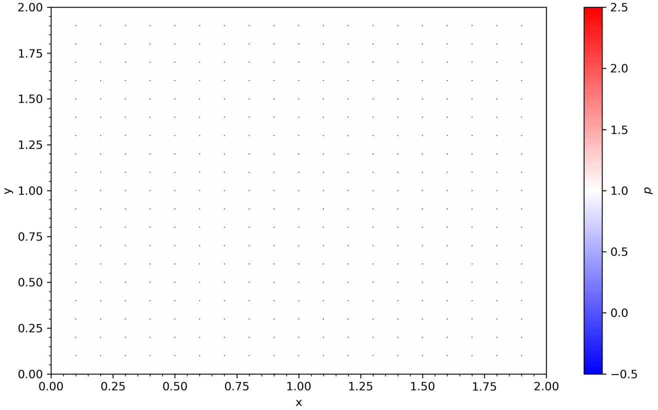

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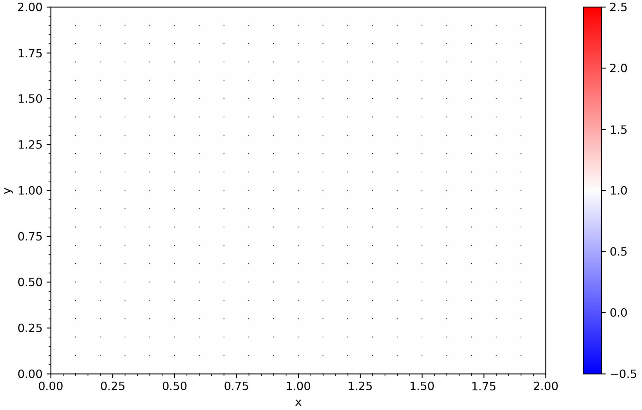

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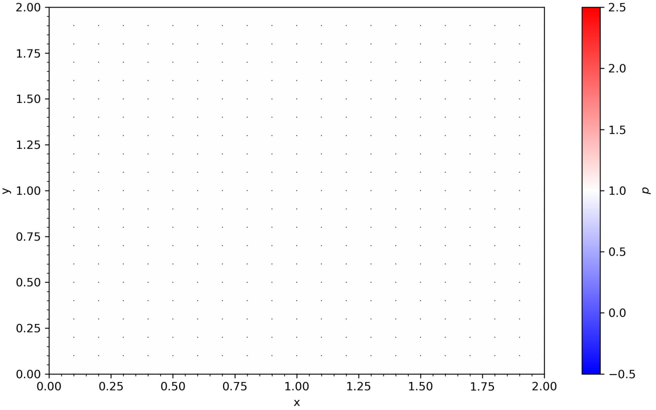

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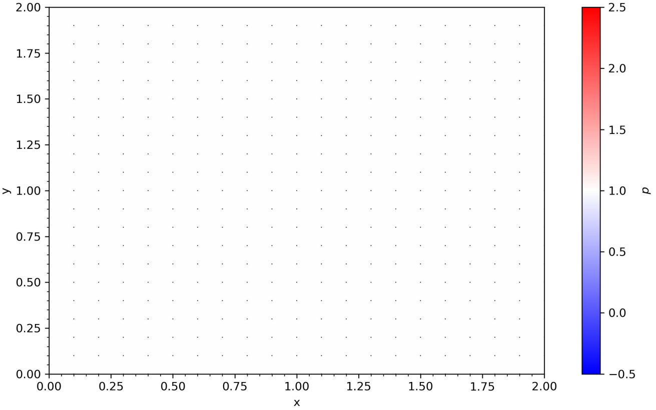

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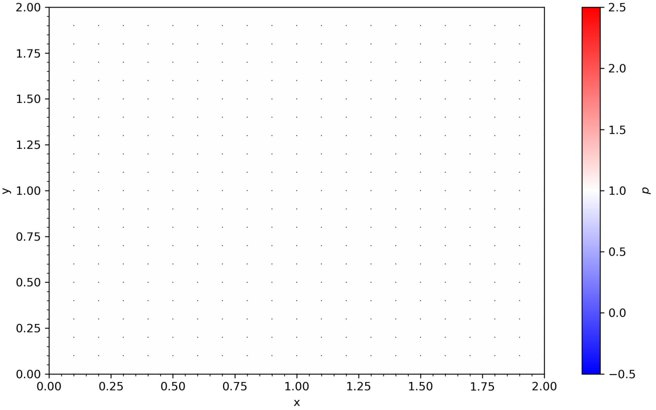

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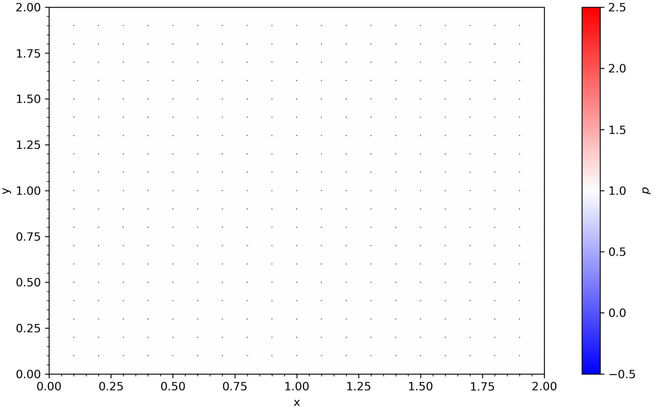

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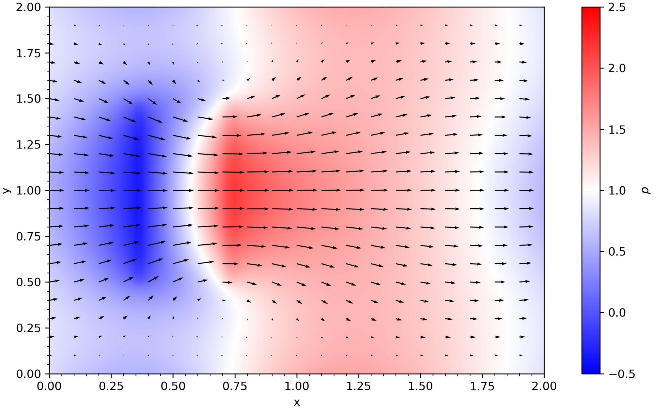

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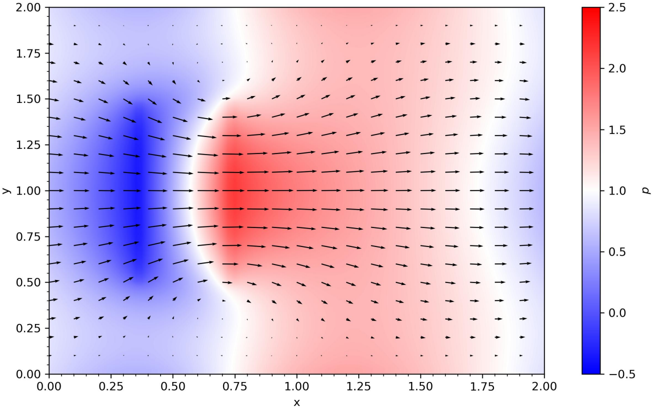

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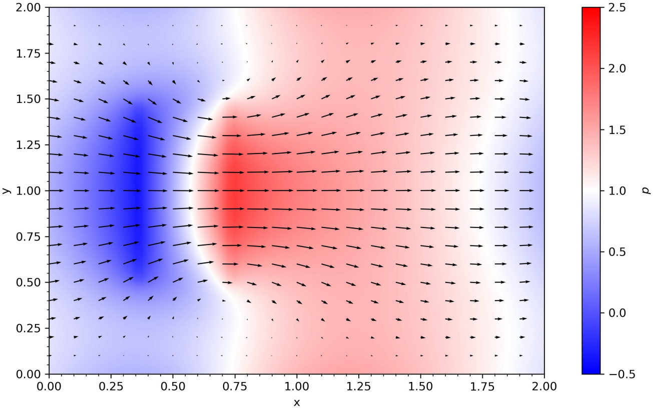

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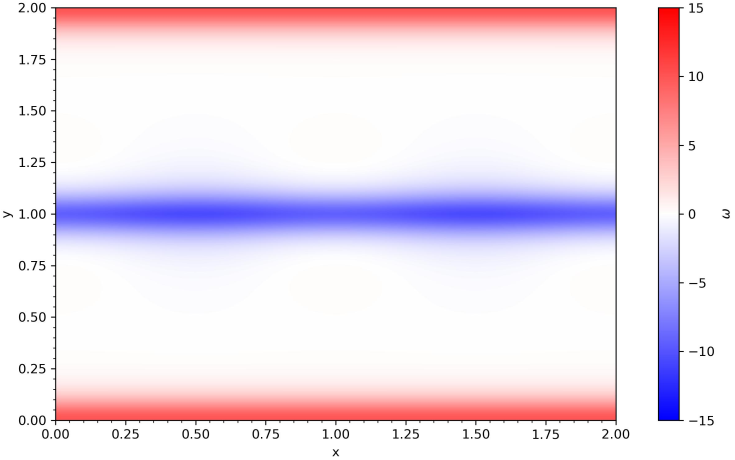

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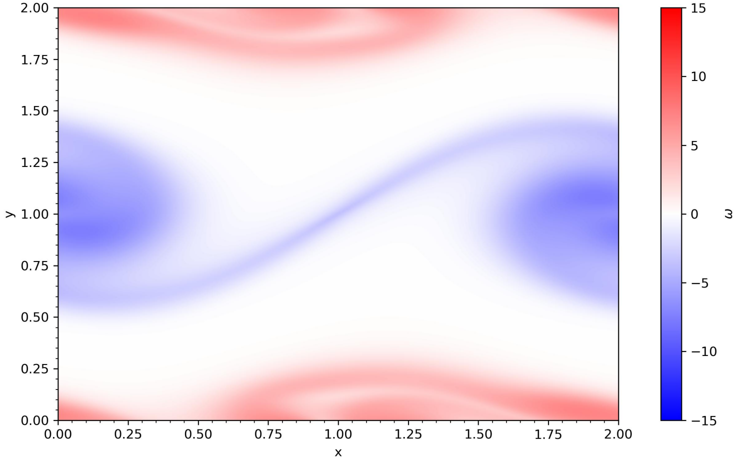

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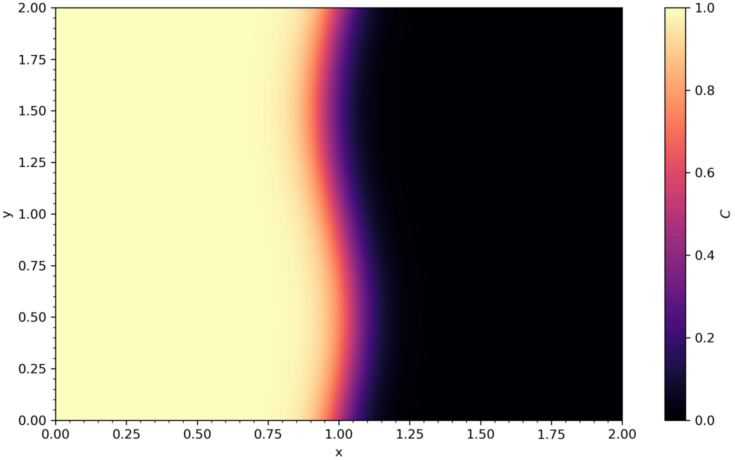

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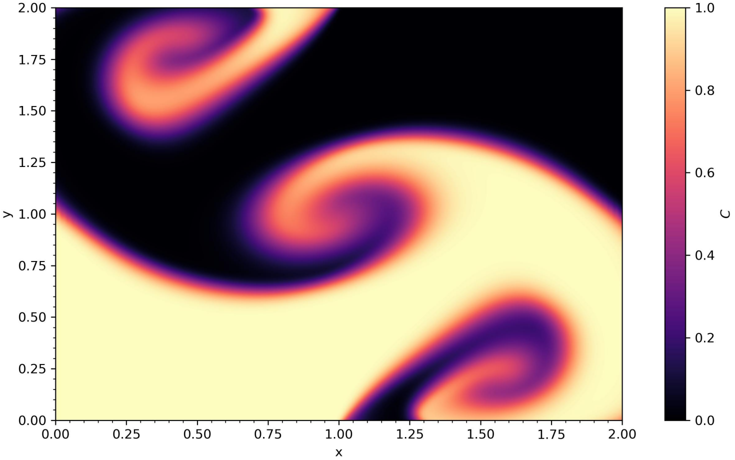

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Thank You!

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- P. Mocz, *Medium*[2] 2020-2025

Resources on GitHub: here